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Psychological phenotypes of sense of coherence: How illness perception mediates distress differently across patient subgroups with hematologic malignancies

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ABSTRACT

Background: Sense of coherence (SOC) plays a critical role in how patients with hematologic malignancies cope with disease-related stress. However, the heterogeneity of SOC and its mechanisms influencing anxiety and depression remain unclear. This study aimed to identify latent SOC profiles and to examine the mediating role of illness perception in the SOC–anxiety and depression relationship.

Methods: A cross-sectional study was conducted among 486 inpatients with hematologic malignancies. SOC, illness perception, and symptoms of anxiety and depression were assessed using the SOC-13, the BIPQ-8, and the PHQ-4. Descriptive analyses, Logistic regression, latent profile analysis, network analysis, and mediation analyses were performed to examine SOC profile classification, illness perception with anxiety-depression associations, and their underlying mechanisms.

Results: Latent profile analysis identified three SOC subgroups: Low SOC–Psychologically Vulnerable (26.8%), Moderate SOC–High Control Low Understanding (42.2%), and High SOC–Resilient Adaptive (31.1%), with age, residence, diagnosis, chronic disease, and social support as significant predictors of profile membership. Network analysis revealed that tension/anxiety (D1) and low mood (D3) were central symptoms across SOC profiles, while emotional representation (P8) consistently emerged as a key bridge node linking illness perception with emotional distress. Mediation analysis further showed that illness perception significantly mediated the SOC–anxiety and depression link, with the strongest effect observed in Profile 2, weaker in Profile 1, and non-significant in Profile 3.

Conclusions: SOC shows heterogeneous profiles with distinct structures. Illness perception partially mediates the SOC–anxiety/depression link, moderated by SOC profiles, highlighting the need for tailored psychological interventions.

1. Introduction

Hematologic malignancies (Mitchell et al., 2011; Tseng & Ng, 2020) represent a major global public health issue, with recent estimates indicating over 1 million prevalent cases worldwide among children, adolescents, and young adults in 2021 (Xu et al., 2026). Unlike patients with many solid tumors, hematologic malignancies confront a unique constellation of stressors, including exceptionally long treatment cycles, chronic uncertainty, prolonged immunosuppression, and a persistent fear of relapse and infection, placing patients at heightened risk for severe anxiety and depression (Kuczmarski et al., 2024; Ma et al., 2025; Xu et al., 2026). A recent systematic review reported that 31.6% of patients with hematologic malignancies experience depression or adjustment disorders (Mitchell et al., 2011), while 38.2% present with broader emotional disturbances (Kuczmarski et al., 2024; Mitchell et al., 2011). Elucidating the protective factors that facilitate psychological

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adaptation in this population, and clarifying the mechanisms through which they exert their effects, is therefore of critical importance for advancing patient care and improving mental health outcomes.

Among the diverse psychological resources identified in health research, Antonovsky's construct of the Sense of Coherence (SOC) (Li et al., 2023; Moksnes, 2021) stands out as a central determinant of resilience. SOC represents a pervasive, enduring, and dynamic orientation of confidence that life's challenges are comprehensible, manageable, and meaningful (Li et al., 2023; Liu, Wang, et al., 2025; Moksnes, 2021; Vähäaho et al., 2021). Extensive evidence demonstrates that SOC serves as a critical protective factor against psychosocial stress, exerting a strong inverse association with anxiety and depression (Li et al., 2023). However, the role of SOC as a reliable predictor of physical health outcomes has been a subject of scholarly debate. Seminal reviews have suggested that while the link between SOC and mental health is robust and consistent, its association with physical health indicators is often weaker and less conclusive (Flensborg-Madsen et al., 2005; Galletta et al., 2019). This distinction provides a strong theoretical justification for focusing specifically on mental health indicators, as SOC's primary influence likely operates through psychological adaptation pathways (Galletta et al., 2019). Yet, existing studies in patients with hematologic malignancies have largely conceptualized SOC as a homogeneous, continuous variable, overlooking potential latent subgroups within this population (Vähäaho et al., 2021). Such an average effect obscures clinically meaningful heterogeneity, limiting our understanding of patients' diverse psychological trajectories (Vähäaho et al., 2021).

To establish a robust theoretical foundation for our mediation model, we draw upon the transactional Stress and Coping Theory. This framework posits that psychological distress is not a direct result of a stressor—such as a cancer diagnosis—but is mediated by an individual's cognitive appraisal of the event and their available coping resources. Within this model, we conceptualize SOC as a critical personal coping resource. According to Antonovsky's theory (Moksnes, 2021), individuals with high SOC possess a view that enables them to mobilize resources effectively for functional coping and emotional regulation (Moksnes, 2021).

Illness perception (Ma et al., 2024), which refers to patients' cognitive and emotional representations of their disease, functions as the primary cognitive appraisal process in this context (Xiong et al., 2023). The Stress and Coping Theory suggests that personal resources like SOC directly shape these appraisals. Specifically, individuals with a stronger SOC are more likely to appraise their illness as comprehensible and controllable, leading to more adaptive illness perceptions (Zamanian et al., 2021). Conversely, those with a weaker SOC may adopt catastrophic, uncontrollable appraisals, which in turn exacerbate anxiety and depression (Moksnes, 2021; Zamanian et al., 2021). Therefore, we propose that illness perception acts as a pivotal mediator, translating the protective effects of the SOC resource into specific cognitive appraisals that ultimately determine emotional outcomes. Despite the strong theoretical basis, rigorous empirical validation in the unique context of hematologic malignancies is lacking.

To further dissect the complex interactions within this framework, our study adopts an integrative strategy combining person- and variable-centered approaches. While traditional regression is limited, network analysis (Fried et al., 2016) offers a nuanced view of the links between illness perception and anxiety and depression, and can identify key bridge symptoms for targeted interventions (Zhu et al., 2023). Latent profile analysis (Ma et al., 2025) will first identify heterogeneous SOC subgroups among hematologic cancer patients. We will then construct weighted networks for each subgroup to compare their structural properties (Ma et al., 2025). Finally, moderated mediation models will test the mediating role of illness perception and the moderating effect of SOC subgroup membership (Tang et al., 2013). Based on this integrated theoretical and methodological framework, we formally hypothesize that: (H1) patients with hematologic malignancies exhibit distinct latent SOC profiles; (H2) the structure of the illness perception–anxiety–depression network varies across SOC subgroups; and (H3) illness perception mediates the relationship between SOC and psychological distress, with the strength of this mediation effect being moderated by SOC profile membership.

Although some of the relationships among these variables have been widely documented in general oncology populations, patients with hematologic malignancies face prolonged treatment, chronic immunosuppression, and persistent relapse uncertainty, which may reshape psychological adaptation processes. It therefore remains unclear whether existing models can be directly generalized to this group. This study aims to identify latent SOC profiles and their heterogeneity in patients with hematologic malignancies, examine subgroup-specific differences in the illness perception, anxiety, and depression network, and quantify the mediating role of illness perception and its moderation by SOC profile. The findings will provide a theoretical basis and practical guidance for developing tailored psychological interventions for distinct SOC subgroups.

2. Methods

2.1. Participants

This cross-sectional study was conducted between July and October 2023 using a convenience sampling method. Participants were patients from the hematology departments of four tertiary hospitals in Hunan Province in China. Inclusion criteria were: (1) pathologically confirmed hematologic malignancy; (2) currently receiving inpatient treatment; and (3) adequate communication skills. Exclusion criteria included: (1) cognitive impairment or diagnosis of a psychiatric disorder; and (2) comorbid severe diseases of vital organs, such as the heart, brain, or kidneys.

2.2. Measures

2.2.1. Sociodemographic and clinical characteristics

Based on a literature review (Hinz et al., 2023; Liu, Wang, et al., 2025; Liu, Zhang, et al., 2025; Schäfer et al., 2023; Vähäaho et al., 2021; Zamanian et al., 2021) and team discussions, the research group developed a self-designed demographic and clinical questionnaire. The questionnaire collected 15 variables, including age, sex, marriage status, educational level, occupation, residence, family monthly income, diagnosis, chronic disease, course of disease, treatment stage, relapse, insurance, complication, and social support.

2.2.2. Sense of Coherence Scale (SOC-13)

The SOC-13 scale was originally developed by Antonovsky (Liu et al., 2024) and has been adapted into Chinese by Bao Leiping et al. (Liu et al., 2024). The SOC-13 consists of 13 items across three dimensions: comprehensibility (5 items: 2, 6, 7, 8, 12), manageability (4 items: 3, 4, 5, 9, 10), and meaningfulness (4 items: 1, 9, 11, 13). Responses are rated on a 7-point Likert scale, with 5 items reverse-scored (Mitchell et al., 2011; Tseng & Ng, 2020; Vähäaho et al., 2021; Xiong et al., 2023; Xu et al., 2026). Total scores range from 13 to 91, with higher scores indicating greater sense of coherence. Scores of 80–91, 64–79, and 13–63 represent high, moderate, and low levels of SOC, respectively. In the present study, the scale demonstrated good internal consistency with a Cronbach's α of 0.865.

2.2.3. The Brief Illness Perception Questionnaire (BIPQ-8) (Galletta et al., 2019)

The Chinese version of the BIPQ-8 (Broadbent et al., 2006), translated by Mei Yaqi et al. (Ma et al., 2024), was used to assess patients' perceptions and emotional responses to their illness. The scale includes nine items: item 1 assesses the impact of the illness on daily life; item 2, the perceived duration of the illness; item 3, perceived personal control; item 4, perceived treatment effectiveness; item 5, perception of

symptoms; item 6, concern about the illness; item 7, understanding of the illness; and item 8, emotional response to the illness. All items are scored on a 0–10 scale. Item 9 is an open-ended question investigating patients' perceived causes of their illness. Total scores range from 0 to 80, with higher scores indicating stronger overall illness perception. In the present study, only the first eight items were analyzed, as the ninth item is open-ended and not suitable for quantitative scoring or statistical analysis. The scale demonstrated excellent internal consistency, with a Cronbach's α of 0.903.

2.2.4. Patient Health Questionnaire-4 (PHQ-4)

Anxiety and depression were measured using the PHQ-4 (Ma et al., 2025). Developed by Kroenke et al., the PHQ-4 combines the first two items of the PHQ-9 (PHQ-2, assessing depressive symptoms) and the first two items of the GAD-7 (GAD-2, assessing anxiety symptoms) to evaluate patients' emotional states over the past two weeks. Each item is rated on a 4-point scale: “not at all” (0), “several days” (1), “more than half the days” (2), and “nearly every day” (3), yielding a total score range of 0–12. Scores of 0–2 indicate no anxiety or depression, 3–5 mild, 6–8 moderate, and 9–12 severe anxiety or depression. In the present study, the PHQ-4 demonstrated good internal consistency (Cronbach's α = 0.84).

2.2.5. Perceived Social Support Scale (PSSS)

Originally developed by Zimet et al. (Zimet et al., 1990) and adapted into Chinese by Jiang Qianjin (Ma et al., 2025), the scale assesses individuals' perceived support from multiple sources, including family, relatives, colleagues, and friends. The PSSS comprises 12 items rated on a 7-point Likert scale ranging from “strongly disagree” (0) to “strongly agree” (7), yielding total scores from 12 to 84. Higher scores indicate better perceived social support. Scores $\leq$36 indicate low support, 37–60 moderate support, and 61–84 high support (Zimet et al., 1990). In the present study, Cronbach's α was 0.901.

2.3. Statistical analysis

SPSS 24.0, Mplus 8.3, PROCESS Macro 4.2 and JASP 0.18.2 were used for data analysis. A two-tailed P value <0.05 was considered statistically significant.

(1) Descriptive analyses: Continuous variables presented as mean ± standard deviation and categorical variables as frequencies and percentages. Chi-square tests or Fisher's Exact Test were used to compare demographic characteristics and scale scores across SOC subgroups, identifying variables with statistically significant differences. Logistic regression analysis was then performed to determine predictors of SOC classification.

(2) Latent profile analysis (LPA) (Liu, Zhang, et al., 2025; Ma et al., 2025): The 13 items of the SOC-13 scale reported by patients with hematologic malignancies were used as observed variables. Models with one to five classes were explored, and the optimal class solution was selected based on a combination of fit indices and practical interpretability. A latent profile plot was subsequently generated. The specific model fit indices were as follows: (i) information criteria: Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and sample-size adjusted BIC (aBIC), with lower values indicating better model fit; (ii) classification accuracy: entropy, ranging from 0 to 1, with values closer to 1 indicating more precise classification; (iii) likelihood ratio tests: the Lo-Mendell-Rubin likelihood ratio test (LMR) and the bootstrapped likelihood ratio test (BLRT), with $P < 0.05$ indicating that the model with the additional class provides a significantly better fit than the model with one fewer class. These indices serve as references, and interpretability of the classes was also considered when determining the optimal model.

(3) Network Analysis (Zhu et al., 2023): To investigate the association structure between illness perception and anxiety–depression, a weighted undirected network was constructed using R 4.3.0. Nodes included dimensions of illness perception and anxiety–depression symptoms (e.g., D1, D2, D3, P1, P2), while edges represented partial correlations. A graphical LASSO (Least Absolute Shrinkage and Selection Operator) was applied to estimate a sparse network. Network visualization was performed using the Fruchterman–Reingold algorithm. Network metrics, including node strength and bridge centrality, were used to identify key nodes linking illness perception and anxiety–depression. Analyses were conducted for the overall sample and for different SOC subgroups, allowing comparison of network structures to explore patterns of illness perception–anxiety–depression associations across SOC profiles. The stability of centrality indices was evaluated using a case-dropping bootstrap procedure, and correlation stability (CS) coefficients were calculated to assess the robustness of the network estimates.

(4) Mediation Analysis (Tang et al., 2013): R 4.3.0 was used to further examine the mechanisms by which SOC influences anxiety–depression and the mediating role of illness perception, employing a moderated mediation model. The independent variable was SOC profile (either as a continuous measure for the overall sample or as latent profile categories), the mediator was the illness perception score, and the dependent variable was anxiety–depression level. Path coefficients were estimated using partial least squares, and indirect effects with 95% confidence intervals were computed via bootstrapping with 5000 resamples. Analyses were conducted for both the overall sample and SOC subgroups to compare differences in mediation effects and clarify the moderating role of SOC profile on the mediation pathway.

2.4. Sample size

To assess potential common method bias, Harman's single-factor test was conducted. All measurement items were entered into an exploratory factor analysis without rotation to examine the variance explained by the first unrotated factor. For logistic regression, the sample size was estimated based on the rule of events per variable (EPV $\geq$10) (van Smeden et al., 2019), and accounting for approximately 20% invalid questionnaires, at least 180 participants were required. For LPA, a minimum of 300 participants was recommended to ensure sufficient statistical power and reliable identification of small subgroups (Ma et al., 2025). For network analysis (Fried et al., 2016), considering the number of nodes and potential edges, the sample size is typically required to be 5–10 times the number of nodes to ensure accurate estimation of network structure and centrality metrics. For mediation analysis (Fritz et al., 2015), according to Fritz & MacKinnon's guidelines, at least 80–100 participants are needed to achieve adequate statistical power. Taken together, each latent profile should include more than 80 participants, and the overall sample size should exceed 300 participants to satisfy all analytic requirements.

2.5. Ethical consideration

A portion of the data used in this study were derived from the same dataset as a previously published study (Ma et al., 2025). While the data source is shared, this study addresses distinct research questions and employs new analyses not reported in the prior publication. This study was approved by the Medical Ethics Committee of the xx (Approval No.: 2024-S425). Prior to enrollment, all participants provided written informed consent after receiving a detailed explanation of the study's objectives, procedures, potential risks, and anticipated benefits. Participants were informed of their right to withdraw from the study at any time without any impact on their clinical care or treatment. All data collected were anonymized using unique numerical identifiers and

securely stored on a password-protected computer accessible only to the research team. This cross-sectional study was conducted in accordance with the Declaration of Helsinki.

3. Results

3.1. Sociodemographic and clinical characteristics of the participants

The first unrotated factor accounted for 13.92% of the total variance, which is below the commonly recommended threshold of 50%, suggesting that common method bias did not affect the results. A total of 486 valid questionnaires were collected. The largest proportion of participants (41.36%) were aged 40 to 59 years. Most had completed junior high school education. Nearly half of the patients (42.59%) had previously worked as laborers, and the vast majority (79.01%) were married. Monthly income data indicated that over one-third of participants (38.48%) reported a household income between 5000 and 10,000 RMB (1 RMB = 0.14 USD). Other demographic characteristics are presented in Table 1.

Table 1
Sociodemographic and clinical characteristics of the participants.

Category		All [Case (Percentage, %)]	C1 [Case (Percentage, %)]	C2 [Case (Percentage, %)]	C3 [Case (Percentage, %)]	χ^2/H	*P*
Age (years)	0–19	21 (4.32)	3 (2.31)	9 (4.39)	9 (5.96)	23.706a	0.003
	20–39	93 (19.14)	15 (11.54)	55 (26.83)	23 (15.23)		
	40–59	201 (41.36)	53 (40.77)	87 (42.44)	61 (40.40)		
	60–79	131 (26.95)	45 (34.62)	39 (19.02)	47 (31.13)		
	80+	40 (8.23)	14 (10.77)	15 (7.32)	11 (7.28)		
Sex	Male	262 (53.91)	73 (56.15)	109 (53.17)	80 (52.98)	0.361	0.835
	Female	224 (46.09)	57 (43.85)	96 (46.83)	71 (47.02)		
Marriage status	Unmarried	62 (12.76)	8 (6.15)	34 (16.59)	20 (13.25)	9.706	0.138
	Married	384 (79.01)	110 (84.62)	157 (76.59)	117 (77.48)		
	Divorced	21 (4.32)	5 (3.85)	9 (4.39)	7 (4.64)		
	Widowed	19 (3.91)	7 (5.38)	5 (2.44)	7 (4.64)		
Educational level	Primary school and below	99 (20.37)	20 (15.38)	43 (20.98)	36 (23.84)	4.468	0.614
	Junior high school	167 (34.36)	49 (37.69)	65 (31.71)	53 (35.100)		
	Senior high school	94 (19.34)	26 (20.00)	40 (19.51)	28 (18.54)		
	College degree or above	126 (25.93)	35 (26.92)	57 (27.80)	34 (22.52)		
Occupation	Laborers	207 (42.59)	50 (38.46)	84 (40.98)	73 (48.34)	7.248	0.299
	Enterprise	69 (14.20)	17 (13.08)	32 (15.61)	20 (13.25)		
	Civil servant/public institution	43 (8.85)	15 (11.54)	13 (6.34)	15 (9.93)		
	Individual/Retired/Unemployed	167 (34.36)	48 (36.92)	76 (37.07)	43 (28.48)		
Residence	Rural	208 (42.80)	75 (57.69)	80 (39.02)	53 (35.10)	17.208a	0.002
	Township	74 (15.23)	13 (10.00)	36 (17.56)	25 (16.56)		
	City	204 (41.98)	42 (32.31)	89 (43.41)	73 (48.34)		
Family monthly income (Yuan)	>10,000	140 (28.81)	52 (40.00)	49 (23.90)	39 (25.83)	11.349a	0.023
	5000–10,000	187 (38.48)	43 (33.08)	86 (41.95)	58 (38.41)		
	<5000	159 (32.72)	35 (26.92)	70 (34.15)	54 (35.76)		
Diagnosis	Multiple myeloma	167 (34.36)	55 (42.31)	69 (33.66)	43 (28.48)	17.304a	0.008
	Lymphoma	139 (28.60)	34 (26.15)	49 (23.90)	56 (37.09)		
	Leukemia	164 (33.74)	34 (26.15)	83 (40.49)	47 (31.13)		
	Others	16 (3.29)	7 (5.38)	4 (1.95)	5 (3.31)		
Chronic disease	None	358 (73.67)	82 (63.08)	151 (73.66)	125 (82.78)	13.981a	<0.001
	Yes (Hypertension, diabetes, coronary heart disease, chronic obstructive pulmonary disease)	128 (26.33)	48 (36.92)	54 (26.34)	26 (17.22)		
Course of disease	1–6 months	249 (51.23)	65 (50.00)	102 (49.76)	82 (54.30)	8.985	0.176
	6–12 months	99 (20.37)	25 (19.23)	44 (21.46)	30 (19.87)		
	1–3 years	104 (21.40)	24 (18.46)	47 (22.93)	33 (21.85)		
	3+ years	34 (7.00)	16 (12.31)	12 (5.85)	6 (3.97)		
Treatment stage	CAR-T cell therapy	15 (3.09)	4 (3.08)	5 (2.44)	6 (3.97)	9.661	0.467
	Maintenance	73 (15.02)	28 (21.54)	28 (13.66)	17 (11.26)		
	Post-transplantation	23 (4.73)	5 (3.85)	11 (5.37)	7 (4.64)		
	Before-transplantation	64 (13.17)	18 (13.85)	26 (12.68)	20 (13.25)		
	Post-chemotherapy	258 (53.09)	64 (49.23)	115 (56.10)	79 (52.32)		
	Before-chemotherapy	53 (10.91)	11 (8.46)	20 (9.76)	22 (14.57)		
Relapse	Yes	83 (17.08)	23 (17.69)	38 (18.54)	22 (14.57)	1.014	0.602
	None	403 (82.92)	107 (82.31)	167 (81.46)	129 (85.43)		
Insurance	Social medical insurance (Provincial, municipal and urban medical insurance)	146 (30.04)	36 (27.69)	65 (31.71)	45 (29.80)	4.479	0.612
	New Rural Co-operative Medical System	133 (27.37)	42 (32.31)	56 (27.32)	35 (23.18)		
	Commercial insurance/public medical insurance	147 (30.25)	37 (28.46)	57 (27.80)	53 (35.10)		
	Self-paying	60 (12.35)	15 (11.54)	27 (13.17)	18 (11.92)		
Complication	None	353 (72.63)	95 (73.08)	154 (75.12)	104 (68.87)	1.725	0.422
	Yes	133 (27.37)	35 (26.92)	51 (24.88)	47 (31.13)		
Social support	Low	97 (19.96)	18 (13.85)	46 (22.44)	33 (21.85)	18.937	<0.001
	Medium	175 (36.01)	34 (26.15)	84 (40.98)	57 (37.75)		
	High	214 (44.03)	78 (60.00)	75 (36.59)	61 (40.40)		

Note: a, Fisher's Exact Test.
C1: Low SOC–Psychologically Vulnerable Group (n = 130, 26.75%), C2: Moderate SOC–High Control Low Understanding Group (*n* = 205, 42.18%), and C3: High SOC–Resilient Adaptive Group (*n* = 151, 31.07%).

3.2. LPA for SOC in hematologic malignancies

Latent profile analysis showed that AIC, BIC, and aBIC decreased with additional classes, and entropy remained high. LMR and BLRT supported models up to four classes, but the five-class model added minimal statistical improvement and included very small classes with proportions of 0.14 and 0.16, which limited interpretability. Considering model fit, classification accuracy, and clinical relevance, the three-class solution with balanced proportions of 0.27, 0.42, and 0.31 and high entropy was optimal and selected as the latent structure of sense of coherence in hematologic malignancy patients (Fig. 1, Table 2).

Based on the proportions of each item, they were labeled as Profile 1: Low SOC–Psychologically Vulnerable Group (n = 130, 26.75%), Profile 2: Moderate SOC–High Control Low Understanding Group (n = 205, 42.18%), and Profile 3: High SOC–Resilient Adaptive Group (n = 151, 31.07%). Chi-square tests and Fisher's exact tests indicated significant differences among SOC profiles in age, diagnosis, family monthly income, residence, chronic disease, and social support ($P < 0.05$), as shown in Table 1.

Variables with significant differences in univariate analyses (VIF < 5) were included as independent variables, and the three SOC profiles identified by LPA served as the dependent variable. Using the Low SOC profile as the reference, residence, chronic disease, and social support significantly predicted membership in the Moderate SOC profile and High SOC profile. Using the moderate SOC profile as the reference, age, residence, diagnosis, and social support significantly predicted membership in the high SOC profile. Results are presented in Table 3.

3.3. Network analysis results

The case-dropping bootstrap analysis showed that the CS-coefficients for strength centrality were 0.362 (Profile 1), 0.751 (Profile 2), and 0.675 (Profile 3), indicating acceptable to excellent stability of the networks. Network models of illness perception (P1–P8) and anxiety–depression symptoms (D1–D4) were constructed for each SOC profile and the total sample (Fig. 2). In the low SOC profile, the network (12 nodes, 25 edges, sparsity = 0.621) showed D3 (low mood; Strength = 1.945, EI = 1.979, Closeness = 0.750) as a relatively central node, while P8 (emotional representation; Betweenness = 2.148, Closeness = 1.090) appeared to act as a bridge node linked with D1 and D3, suggesting that emotional illness perception may be associated with heightened distress. In the moderate SOC profile (29 edges, sparsity = 0.561), D1 (tension and anxiety; Strength = 2.316, EI = 2.147, Closeness = 1.128) appeared to play a dominant role, with D3 secondary (Strength = 1.013, EI = 1.017). P8 (Strength = 0.607, EI = 0.665) was also connected to D1, indicating a potential link between emotional perception and anxiety-related symptoms. In the high SOC profile (34 edges, sparsity = 0.485), D1 (Strength = 2.551, EI = 2.481, Closeness = 1.686, Betweenness = 2.553) appeared to be one of the most central nodes in the network. P5 (timeline perception; Strength = 0.588, Closeness = 0.733) showed connections with several nodes in the network, while P8 remained linked with depressive symptoms. Overall, emotional representation (P8) consistently appeared as an important connector in the illness perception–anxiety–depression network across SOC profiles (Fig. 3; Table 4).

3.4. Moderated mediation analysis

3.4.1. Correlation analysis

Results showed that SOC was negatively correlated with illness perception and anxiety–depression ($r < 0$, $P < 0.01$), while illness perception was positively correlated with anxiety–depression ($r > 0$, $P < 0.01$) (Table 5).

3.4.2. Mediation analysis

After controlling for covariates, mediation models were tested with anxiety–depression as the dependent variable, SOC as the independent variable, and illness perception as the mediator across All, Profile 1, Profile 2, and Profile 3. In the All participants and Profile 2, illness perception exerted a significant mediating effect on the SOC–distress relationship, while the direct effect of SOC on anxiety–depression also remained significant. This indicates that SOC influenced distress both directly and indirectly through illness perception. However, subgroup heterogeneity was observed: the mediation effect was most pronounced in Class 2, marginal in Class 1, and non-significant in Class 3 (Table 6, Fig. 4).

4. Discussion

This study is the first to integrate latent profile analysis (LPA), network analysis, and mediation analysis to reveal the heterogeneous structure of SOC and its differential mechanisms in influencing anxiety and depression through illness perception.

LPA revealed three SOC profiles among patients with hematologic malignancies: “low coherence–psychologically vulnerable,” “moderate SOC–high manageability/low comprehensibility,” and “high coherence–resilient adaptive.” This typology shows partial concordance with prior studies while also highlighting population-specific characteristics.

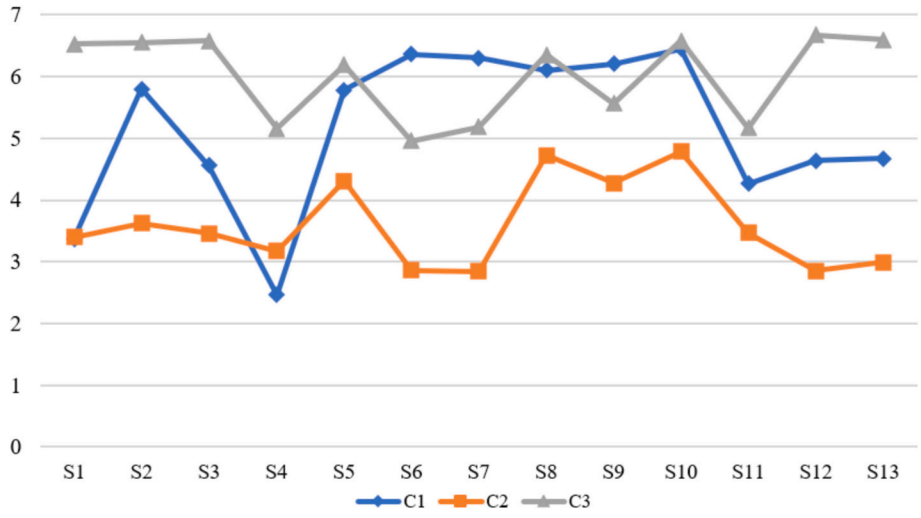


Fig. 1. Latent profile analysis of SOC in patients with hematological malignancies.

Table 2
Latent profile analysis of SOC in patients with hematological malignancies ($n = 486$).

Class	AIC	BIC	aBIC	Entropy	LMR (*P*)	BLRT (*P*)	Proportion
1	25,706.86	25,815.7	25,733.18	–	–	–	–
2	23,857.38	24,024.83	23,897.87	0.976	0.000	0.000	0.43/0.57
3	23,239.61	23,465.67	23,294.27	0.952	0.000	0.000	0.27/0.42/0.31
4	22,869.64	23,154.30	22,938.47	0.937	0.009	0.000	0.18/0.26/0.29/0.27
5	22,603.83	22,947.10	22,686.84	0.957	0.042	0.000	0.18/0.25/0.14/0.27/0.16

Note: AIC, Akaike Information Criterion; BIC, Bayesian Information Criterion; aBIC, adjusted Bayesian Information Criterion; LMRT, Lo-Mendell-Rubin Test; BLRT, Bootstrap Likelihood Ratio Test.

Table 3
Results of multivariate analysis.

Reference group		C3						C2		
Comparison group		C1[b]			C2			C1		
		OR	95% CI	*P*	OR	95% CI	*P*	OR	95% CI	*P*
Age (years)	0–19	0.202	0.040–1.034	0.055	0.705	0.202–2.465	0.584	0.287	0.059–1.400	0.122
	20–39	0.639	0.212–1.923	0.426	2.086	0.812–5.360	0.127	0.306	0.113–0.829	0.020
	40–59	0.782	0.306–1.996	0.606	1.172	0.495–2.778	0.718	0.667	0.282–1.574	0.355
	60–79	0.776	0.291–2.065	0.611	0.615	0.244–1.550	0.303	1.260	0.503–3.156	0.621
	80+	0[b]								
Residence	Rural	2.985	1.653–5.391	0.000	1.243	0.742–2.082	0.409	2.402	1.371–4.208	0.002
	Township	1.318	0.575–3.023	0.514	1.038	0.548–1.966	0.909	1.270	0.573–2.814	0.556
	City	0[b]								
Family monthly income (Yuan)	>10,000	1.642	0.844–3.195	0.144	0.972	0.534–1.770	0.927	1.689	0.894–3.191	0.106
	5000–10,000	1.089	0.571–2.075	0.796	1.065	0.633–1.794	0.812	1.022	0.554–1.885	0.944
	<5000	0[b]								
Diagnosis	Multiple myeloma	0.614	0.161–2.346	0.476	2.520	0.613–10.362	0.200	0.244	0.059–1.006	0.051
	Lymphoma	0.297	0.077–1.149	0.079	1.205	0.295–4.924	0.795	0.247	0.058–1.043	0.057
	Leukemia	0.389	0.099–1.524	0.175	2.339	0.571–9.577	0.238	0.166	0.039–0.702	0.015
	Others	0[b]								
Chronic disease	None	0.321	0.176–0.585	0.000	0.469	0.269–0.817	0.008	0.685	0.404–1.160	0.159
	Yes	0[b]								
Social support	Low	0.418	0.205–0.853	0.017	1.275	0.705–2.306	0.421	0.328	0.167–0.645	0.001
	Medium	0.415	0.232–0.743	0.003	1.241	0.750–2.053	0.402	0.335	0.193–0.580	0.000
	High	0[b]								

C1: Low SOC–Psychologically Vulnerable Group (n = 130, 26.75%), C2: Moderate SOC–High Control Low Understanding Group (n = 205, 42.18%), and C3: High SOC–Resilient Adaptive Group (n = 151, 31.07%). b Reference.

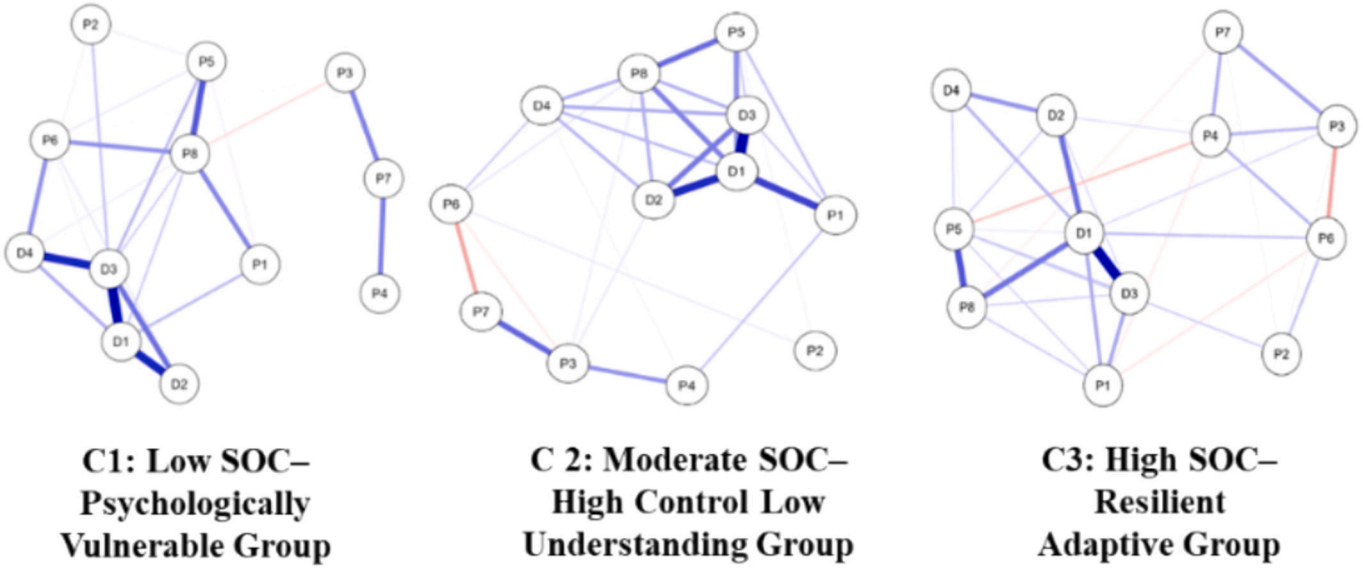


Fig. 2. Network figures of "illness perception" and "anxiety-depression" in patients with hematological malignancies (D1–D4: D1, feeling tense, anxious, or restless; D2, inability to stop or control worrying; D3, feeling down, depressed, or hopeless; D4, feeling little interest or motivation when doing things. P1-P8: P1, assesses the impact of the illness on daily life; P2, the perceived duration of the illness; P3, perceived personal control; P4, perceived treatment effectiveness; P5, perception of symptoms; P6, concern about the illness; P7, understanding of the illness; P8, emotional response to the illness).

For example, Liu et al. (Liu et al., 2024) similarly identified three SOC profiles among patients undergoing chemotherapy and radiotherapy:

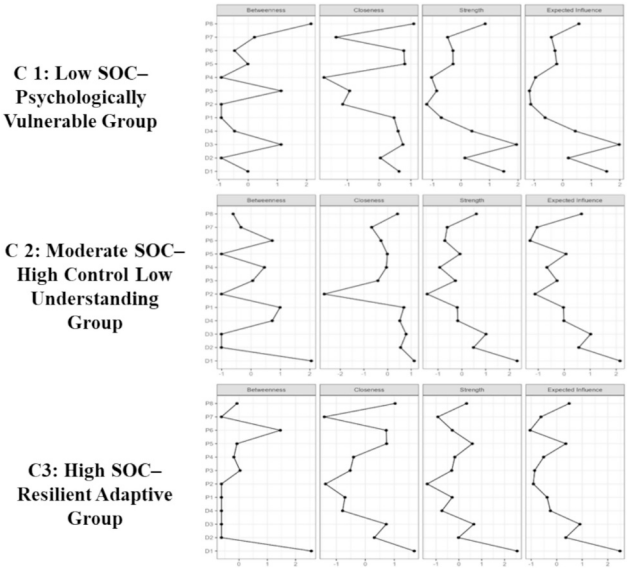


Fig. 3. Centrality indices (betweenness, closeness, strength, and expected influence) of illness perception and anxiety-depression nodes in overall and SOC subgroups.

Table 4
Comparison of Node Centrality Index (Strength) in Networks in overall and subgroups.

Variable	Low SOC–Psychologically Vulnerable Group	Moderate SOC–High Control Low Understanding Group	High SOC–Resilient Adaptive Group
D1	1.497	2.316	2.551
D2	0.136	0.489	−0.014
D3	1.945	1.013	0.652
D4	0.382	−0.173	−0.741
P1	−0.691	−0.194	−0.293
P2	−1.202	−1.454	−1.385
P3	−0.852	−0.272	−0.313
P4	−1.033	−0.927	−0.177
P5	−0.279	−0.081	0.588
P6	−0.282	−0.713	−0.287
P7	−0.469	−0.612	−0.918
P8	0.848	0.607	0.337

Table 5
Correlation analysis of SOC, disease perception, anxiety and depression.

Variables	SOC	Illness perception	Depression-anxiety
SOC	1.000		
Illness perception	−0.263	1.000	
Depression-anxiety	−0.170	0.623	1.000

"Low level–High meaningfulness," "Moderate level–High manageability," and "High level–Balanced", with distribution patterns largely consistent with our findings. However, in our cohort, the moderate profile was characterized by relatively higher manageability but lower comprehensibility, a pattern that may reflect the prolonged treatment cycles and complexity of hematologic malignancies. Under such circumstances, patients may gradually develop a sense of control through daily disease management, while still experiencing difficulties in fully understanding the disease trajectory and its underlying mechanisms. Comparable heterogeneity has also been observed among caregivers of breast cancer patients (Wang et al., 2024), where SOC was categorized into "Low coherence–Meaning," "Moderate coherence–Manageability," and "High coherence–Optimism". Notably, the proportion of high-level SOC in caregivers (54.7%) exceeded that of our patient population

Table 6
Mediating effect analysis of SOC, disease perception, anxiety and depression.

	Effect	SE	LLCI	ULCI	Effect size
All-Total effect	−0.063	0.010	−0.083	−0.042	/
All-Direct effect	−0.028	0.008	−0.055	−0.022	44.44%
All- Indirect effect	−0.024	0.007	−0.037	−0.010	38.10%
C1- Total effect	−0.243	0.024	−0.291	−0.195	/
C1- Direct effect	−0.157	0.023	−0.203	−0.110	64.61%
C1- Indirect effect	−0.086	0.016	−0.119	−0.057	35.39%
C2- Total effect	−0.030	0.034	−0.098	0.038	/
C2- Direct effect	−0.015	0.026	−0.067	0.037	50.00%
C2- Indirect effect	−0.015	0.023	−0.062	0.030	50.00%
C3- Total effect	−0.071	0.047	−0.163	0.021	/
C3- Direct effect	−0.034	0.044	−0.120	0.052	47.89%
C3- Indirect effect	−0.037	0.023	−0.091	−0.002	52.11%

LLCI = Lower Level Confidence Interval; ULCI = Upper Level Confidence Interval.
C1, Low SOC–Psychologically Vulnerable Group; C2, Moderate SOC–High Control Low Understanding Group; C3, High SOC–Resilient Adaptive Group.

(31.07%), suggesting that caregivers, given their supportive role, may rely more heavily on perceived resources, meaning, and optimism to sustain psychological adjustment under stress. In contrast, a study of undergraduate nursing students identified four SOC categories (Liu, Zhang, et al., 2025), including "Low sensitivity–Low coping" and "High sensitivity–High coping". These findings further underscore that factors such as age, psychological maturity, coping experience, and social support may shape the manifestation and structure of SOC across populations.

Further analyses revealed significant differences across SOC profiles in terms of age, residence, household income, diagnosis, residence, chronic disease status, and perceived social support. These findings are broadly consistent with prior evidence while also underscoring population-specific features. Patients aged 20–39 years were more likely to belong to the moderate SOC profile. This age range often coincides with critical family and career responsibilities (Buro et al., 2023; Hinz et al., 2023); individuals typically possess certain coping resources but also face heightened social and economic pressures (Buro et al., 2023), which may explain the "high manageability but low comprehensibility" profile. Schäfer et al. (Schäfer et al., 2023) similarly reported a steady increase in SOC from childhood through young adulthood. Rural patients were disproportionately represented in the low SOC profile, which may reflect limited access to medical resources, weaker health education, and less robust social support systems (Wang et al., 2024). A study of Chinese cancer caregivers (Wang et al., 2024) similarly reported lower SOC among rural populations, highlighting the psychological impact of urban–rural disparities.

Clinical diagnosis further shaped SOC profiles: patients with leukemia tended to cluster in the moderate SOC profile, a pattern that may be related to the protracted and cyclical treatment course and complex disease management requirements (Liu, Wang, et al., 2025), which may foster reliance on manageability while limiting comprehensibility and meaningfulness. In contrast, patients without chronic comorbidities were more likely to fall into the high SOC profile, suggesting that multimorbidity exacerbates uncertainty and burden, thereby reducing adaptive capacity. Jasiński et al. similarly reported that presence of chronic disease was negatively correlated with SOC (Jasiński et al., 2025). Notably, social support varied significantly across profiles. Paradoxically, patients reporting low-to-moderate support were not restricted to the low SOC profile but also appeared in the moderate and even high SOC profiles. This indicates that SOC is shaped not solely by external resources but also by psychological traits, coping strategies, and experiential mastery of disease management (Song et al., 2022). Accordingly, even with limited external support, some individuals may attain high coherence through internal regulation and adaptive resilience (Li et al., 2023). Together, these findings substantiate the sociodemographic and clinical heterogeneity underpinning SOC, and they

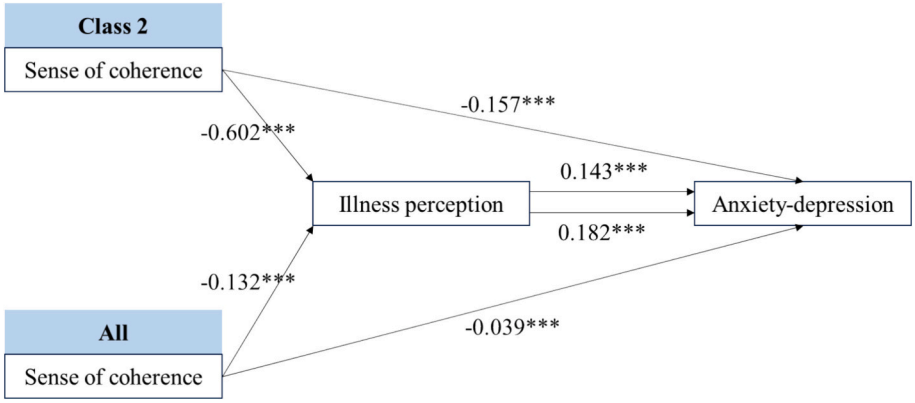


Fig. 4. Mediating effects of illness perception on anxiety-depression (***$P < 0.001$).

underscore the need for differentiated intervention strategies to enhance psychological adaptation among patients with hematologic malignancies.

Network analysis showed that, across all SOC profiles (C1–C3) and the overall sample, D1 (feeling tense or anxious), D2 (uncontrollable worry), D3 (feeling depressed or hopeless), P5 (perception of symptoms), and P8 (emotional response to the illness) consistently emerged as central nodes. This indicates that anxiety, depressive symptoms, and illness-related emotional responses occupy prominent positions in the psychological distress network of patients with hematologic malignancies (Hinz et al., 2023; Kuczmarski et al., 2024). The prominence of these nodes likely reflects the prolonged and intensive treatment trajectories typical of hematologic malignancies, where emotional responses and symptom burden mutually reinforce each other, forming a self-perpetuating cycle (Fang et al., 2024; Kuczmarski et al., 2024). Consistent with prior oncology network studies (Sharpley et al., 2023), core symptoms of anxiety and depression drive overall distress, with disease-related symptom severity and its emotional consequences further amplifying psychological burden (Cai et al., 2024; Zhan et al., 2024). Notably, these central nodes remained stable across SOC profiles, suggesting a robust, cross-population core of psychological distress. These findings emphasize that psychosocial interventions should both address subgroup-specific needs and prioritize alleviating core anxiety and depressive symptoms.

This study further examined the indirect effect of SOC on anxiety and depression via illness perception. The mediation pathway was significant in Class 2 (moderate SOC–high manageability/low comprehensibility) and in the overall sample, but not in Class 1 (low coherence–psychologically vulnerable) or Class 3 (high coherence–resilient adaptation). These findings indicate that the mechanisms linking SOC to anxiety and depression differ across patient subgroups (Song et al., 2022). Specifically, moderate SOC patients typically exhibit strong manageability but limited understanding of their illness, making them more susceptible to anxiety and depression through negative illness perceptions. This aligns with international studies in breast cancer caregivers (Wang et al., 2024) populations, which report that moderate SOC individuals often display a “high control–low comprehension” paradox, rendering them particularly influenced by disease perception. In contrast, low SOC patients possess limited personal resources and coping capacity; their anxiety and depression may arise directly from SOC-related vulnerability, attenuating the mediating role of illness perception (Kuczmarski et al., 2024). For high SOC patients, strong sense of meaningfulness, comprehensibility, and manageability likely buffer the negative impact of illness perception on psychological outcomes, rendering the mediation effect non-significant. These differential patterns highlight the complexity of the relationships among SOC, illness perception, and anxiety/depression, and suggest that clinical interventions should be stratified by SOC profile: for moderate SOC patients, targeting illness perception is critical; for low SOC patients, enhancing resources and coping skills is paramount; and for high SOC patients, maintaining and reinforcing existing psychological resilience is key.

4.1. Study limitations

This study has several limitations that should be addressed in future research. First, the cross-sectional design limits our findings to associations among variables and precludes causal inferences; future longitudinal or interventional studies are needed to clarify the mechanisms linking SOC, illness perception, and anxiety/depression. Second, while the PHQ-4 scale offers brevity and acceptable reliability and validity, it does not comprehensively capture the full spectrum of anxiety and depressive symptoms, potentially underestimating or omitting certain clinical features. Future studies should employ more comprehensive assessment tools. Finally, as the sample was drawn from a single hospital, the generalizability of our findings is limited. Multicenter studies with larger and more diverse populations are warranted to enhance the external validity and applicability of the results.

5. Conclusions

In summary, this study systematically elucidates the heterogeneous structure of SOC and its associations with anxiety and depression among patients with hematologic malignancies. SOC was classified into three latent profiles: low SOC–psychologically vulnerable, moderate SOC–high control low understanding, and high SOC–resilient adaptative. These profiles differed significantly across age, residence, household income, diagnosis, chronic disease, and perceived social support, reflecting population-specific characteristics. Network analysis identified anxiety and depressive core symptoms, as well as symptom burden and emotional impact from illness perception, as stable central nodes, highlighting them as cross-population key targets for intervention. Mediation analysis further revealed that the mechanism linking SOC to anxiety and depression via illness perception varies by subgroup: moderate SOC patients are particularly susceptible to the negative influence of illness perception; in low SOC patients, psychological distress is primarily driven by direct coping deficits; and high SOC patients exhibit robust resilience that partially buffers against adverse effects of illness perception.

Collectively, these findings not only confirm the sociodemographic and clinical heterogeneity underpinning SOC but also provide a theoretical basis for tailored psychosocial interventions: adjusting illness perception for moderate SOC patients, strengthening resources and coping capacity for low SOC patients, and maintaining psychological resilience for high SOC patients, ultimately aiming to enhance overall mental health in patients with hematologic malignancies.

CRediT authorship contribution statement

Guiyuan Ma: Writing – original draft, Visualization, Methodology, Funding acquisition, Data curation, Conceptualization. **Nannan Long:** Writing – original draft, Supervision, Methodology, Conceptualization. **Yuanyuan Li:** Supervision, Methodology, Investigation, Funding acquisition, Formal analysis. **Ping Mao:** Supervision, Software, Methodology, Investigation, Conceptualization. **Chengyuan Li:** Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics statement

The studies involving human participants were reviewed and approved by Ethics Committee of the third Xiangya hospital, Central South University (Approval No.2024-S425). The patients/participants provided their written informed consent to participate in this study.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors declare that no generative AI tools were used in the preparation of this manuscript.

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Declaration of competing interest

The authors declare no conflict of interest.

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Data availability

Data will be made available on request.

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